

Universidad Carlos III de Madrid

Aerospace Engineering

Computer Laboratory Report 2

Particle on Oscillating Loop

Mechanics Applied to Aerospace Engineering

Laboratory Group LA

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1 Introduction

In this laboratory, the motion of a heavy particle P (a bead) that slides without friction on a circular wire of centre C and radius R is studied. The wire lies in the plane Oxy, which rotates about the vertical axis Oy with angular velocity $\Omega(t) = A\omega\sin(\omega t)$, and the position of P is given by the angle φ between $\overrightarrow{CP}$ and the axis Ox (Figure 1).

The main objective is to obtain the equation of motion and to study, with `ode45`, the angle, the reaction of the wire, the inertia forces, the energy and the trajectory of P on the oscillating loop; then the phase portrait of the fixed loop and the critical rate to go over the top; and finally the linearized equation near the stable equilibrium, compared with the full one. Section 2 derives the model and the numerical setup, Section 3 presents and discusses each simulation, Section 4 comments on the model as a whole and Section 5 sums up.

2 Methodology

2.1 Kinematics and degree of freedom

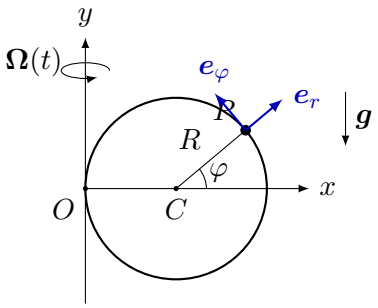


Figure 1: Geometry of the problem. The plane Oxy turns about Oy; $\boldsymbol{k} = \boldsymbol{i} \times \boldsymbol{j}$ points out of the page.

We use two reference frames. $Oxyz$ turns with the plane of the wire: $\boldsymbol{i}$ is horizontal along the diameter through C, $\boldsymbol{j}$ points up along the axis and $\boldsymbol{k} = \boldsymbol{i} \times \boldsymbol{j}$ is normal to the plane. $Ox_1y_1z_1$ is fixed to the ground and coincides with $Oxyz$ at $t = 0$. As shown in Figure 1, the axis Oy touches the wire at O, so $\overrightarrow{OC} = R\,\boldsymbol{i}$.

Degrees of freedom. A free particle has three coordinates. The wire removes two of them, since the bead has to stay in the plane Oxy and at a distance R from C. So there is $3 - 2 = 1$ degree of freedom, and we parametrise it with φ. The plane turns with a known law, so the position of the bead seen from the ground depends on φ and also explicitly on t.

How the plane turns. The angle θ turned by the plane satisfies $\dot\theta = \Omega$ with $\theta(0) = 0$. Integrating with the change $u = \omega\tau$, $\mathrm{d}u = \omega\,\mathrm{d}\tau$, and applying Barrow's rule,

$$\theta(t) = \int_0^t A\omega\sin(\omega\tau)\,\mathrm{d}\tau = A\int_0^{\omega t}\sin u\,\mathrm{d}u = A\big[-\cos u\big]_0^{\omega t} = A(1-\cos\omega t). \tag{1}$$

This means that the plane swings between $\theta = 0$ and $\theta = 2A$. With $A = \pi$ it makes a full turn at $t = T/2$ and then turns back, $T = 2\pi/\omega$ being the period of the plane. The two bases are related by a rotation of angle θ about the vertical axis, and the columns of

$$[{}_1R_0] = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \tag{2}$$

are the components of $\boldsymbol{i}$, $\boldsymbol{j}$, $\boldsymbol{k}$ in $\{\boldsymbol{i}_1, \boldsymbol{j}_1, \boldsymbol{k}_1\}$. These columns are unit vectors perpendicular to each other, so $[{}_1R_0][{}_1R_0]^T = I$, and $\det[{}_1R_0] = \cos^2\theta + \sin^2\theta = 1$. To check it, we differentiate $\boldsymbol{i} = \cos\theta\,\boldsymbol{i}_1 - \sin\theta\,\boldsymbol{k}_1$ and get $\mathrm{d}\boldsymbol{i}/\mathrm{d}t = -\dot\theta(\sin\theta\,\boldsymbol{i}_1 + \cos\theta\,\boldsymbol{k}_1) = -\Omega\,\boldsymbol{k}$, which is exactly $\boldsymbol{\Omega} \times \boldsymbol{i}$.

Motion along the wire. Knowing that the polar unit vectors are $\boldsymbol{e}_r = \cos\varphi\,\boldsymbol{i} + \sin\varphi\,\boldsymbol{j}$ and $\boldsymbol{e}_\varphi = -\sin\varphi\,\boldsymbol{i} + \cos\varphi\,\boldsymbol{j}$, the position of the bead is

$$\boldsymbol{r} = \overrightarrow{OC} + \overrightarrow{CP} = R(1+\cos\varphi)\,\boldsymbol{i} + R\sin\varphi\,\boldsymbol{j}, \qquad x \equiv R(1+\cos\varphi), \tag{3}$$

where x is its distance to the axis of rotation. In $Oxyz$ only φ changes with time. By the chain rule, $\frac{\mathrm{d}}{\mathrm{d}t}[R(1+\cos\varphi)] = -R\dot\varphi\sin\varphi$ and $\frac{\mathrm{d}}{\mathrm{d}t}[R\sin\varphi] = R\dot\varphi\cos\varphi$, so

$$\boldsymbol{v}_{\mathrm{rel}} = -R\dot\varphi\sin\varphi\,\boldsymbol{i} + R\dot\varphi\cos\varphi\,\boldsymbol{j} = R\dot\varphi\,\boldsymbol{e}_\varphi. \tag{4}$$

Differentiating again with the product rule, $\frac{\mathrm{d}}{\mathrm{d}t}(-R\dot\varphi\sin\varphi) = -R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi)$ and $\frac{\mathrm{d}}{\mathrm{d}t}(R\dot\varphi\cos\varphi) = R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)$, so

$$\boldsymbol{a}_{\mathrm{rel}} = -R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi)\,\boldsymbol{i} + R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)\,\boldsymbol{j} = R\ddot\varphi\,\boldsymbol{e}_\varphi - R\dot\varphi^2\,\boldsymbol{e}_r. \tag{5}$$

The first term is the acceleration along the wire and the second one the centripetal acceleration towards C.

Motion seen from the ground. Since O is on the axis and never moves, the velocity and acceleration in the fixed frame are $\boldsymbol{v} = \boldsymbol{v}_{\mathrm{rel}} + \boldsymbol{\Omega}\times\boldsymbol{r}$ and $\boldsymbol{a} = \boldsymbol{a}_{\mathrm{rel}} + \dot{\boldsymbol{\Omega}}\times\boldsymbol{r} + \boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}) + 2\boldsymbol{\Omega}\times\boldsymbol{v}_{\mathrm{rel}}$, with $\Omega = A\omega\sin\omega t$ and $\dot\Omega = A\omega^2\cos\omega t$. Using $\boldsymbol{j}\times\boldsymbol{i} = -\boldsymbol{k}$, $\boldsymbol{j}\times\boldsymbol{j} = \boldsymbol{0}$ and $\boldsymbol{j}\times\boldsymbol{k} = \boldsymbol{i}$, the four extra terms are

$$\boldsymbol{\Omega}\times\boldsymbol{r} = \Omega x\,(\boldsymbol{j}\times\boldsymbol{i}) + \Omega R\sin\varphi\,(\boldsymbol{j}\times\boldsymbol{j}) = -\Omega x\,\boldsymbol{k}, \qquad \dot{\boldsymbol{\Omega}}\times\boldsymbol{r} = -\dot\Omega x\,\boldsymbol{k}, \tag{6}$$

$$\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}) = \Omega\,\boldsymbol{j}\times(-\Omega x\,\boldsymbol{k}) = -\Omega^2 x\,(\boldsymbol{j}\times\boldsymbol{k}) = -\Omega^2 x\,\boldsymbol{i}, \tag{7}$$

$$2\boldsymbol{\Omega}\times\boldsymbol{v}_{\mathrm{rel}} = 2\Omega R\dot\varphi\big[-\sin\varphi\,(\boldsymbol{j}\times\boldsymbol{i}) + \cos\varphi\,(\boldsymbol{j}\times\boldsymbol{j})\big] = 2\Omega R\dot\varphi\sin\varphi\,\boldsymbol{k}. \tag{8}$$

Adding them, we get:

$$\boldsymbol{v} = R\dot\varphi\,\boldsymbol{e}_\varphi - \Omega x\,\boldsymbol{k}, \tag{9}$$

$$\boldsymbol{a} = \big[-R(\ddot\varphi\sin\varphi + \dot\varphi^2\cos\varphi) - \Omega^2 x\big]\boldsymbol{i} + R(\ddot\varphi\cos\varphi - \dot\varphi^2\sin\varphi)\,\boldsymbol{j} + \big[-\dot\Omega x + 2\Omega R\dot\varphi\sin\varphi\big]\boldsymbol{k}. \tag{10}$$

As $\boldsymbol{e}_\varphi \perp \boldsymbol{k}$, the speed is $|\boldsymbol{v}|^2 = R^2\dot\varphi^2 + \Omega^2 x^2$, the speed along the wire plus the speed at which the plane carries the bead around the axis.

2.2 Forces and equation of motion

The only real forces on the bead are its weight and the reaction of the wire. The wire is smooth, so the reaction has no component along its tangent $\boldsymbol{e}_\varphi$:

$$\boldsymbol{W} = -mg\,\boldsymbol{j} = -mg\sin\varphi\,\boldsymbol{e}_r - mg\cos\varphi\,\boldsymbol{e}_\varphi, \qquad \boldsymbol{N} = N_r\,\boldsymbol{e}_r + N_z\,\boldsymbol{k}, \tag{11}$$

where we used $\boldsymbol{j} = \sin\varphi\,\boldsymbol{e}_r + \cos\varphi\,\boldsymbol{e}_\varphi$. Since Newton's law is written in the turning frame, the inertia forces have to be added. From Equations (6) to (8),

$$\boldsymbol{F}_{\mathrm{cf}} = -m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}) = m\Omega^2 x\,\boldsymbol{i} = mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\big[\cos\varphi\,\boldsymbol{e}_r - \sin\varphi\,\boldsymbol{e}_\varphi\big], \tag{12}$$

$$\boldsymbol{F}_{\mathrm{Eul}} = -m\dot{\boldsymbol{\Omega}}\times\boldsymbol{r} = m\dot\Omega x\,\boldsymbol{k} = mA\omega^2 R\cos(\omega t)(1+\cos\varphi)\,\boldsymbol{k}, \tag{13}$$

$$\boldsymbol{F}_{\mathrm{Cor}} = -2m\boldsymbol{\Omega}\times\boldsymbol{v}_{\mathrm{rel}} = -2m\Omega R\dot\varphi\sin\varphi\,\boldsymbol{k} = -2mA\omega R\dot\varphi\sin(\omega t)\sin\varphi\,\boldsymbol{k}, \tag{14}$$

with $\boldsymbol{i} = \cos\varphi\,\boldsymbol{e}_r - \sin\varphi\,\boldsymbol{e}_\varphi$. Newton's second law in $Oxyz$ is then $m\boldsymbol{a}_{\mathrm{rel}} = \boldsymbol{W} + \boldsymbol{N} + \boldsymbol{F}_{\mathrm{cf}} + \boldsymbol{F}_{\mathrm{Eul}} + \boldsymbol{F}_{\mathrm{Cor}}$. The reaction has no component along $\boldsymbol{e}_\varphi$ and the Euler and Coriolis forces point along $\boldsymbol{k}$, so the $\boldsymbol{e}_\varphi$ component has no unknown force, $mR\ddot\varphi = -mg\cos\varphi - m\Omega^2 x\sin\varphi$. And solving for $\ddot\varphi$,

$$\ddot\varphi = -\frac{g}{R}\cos\varphi - A^2\omega^2\sin^2(\omega t)\,(1+\cos\varphi)\sin\varphi. \tag{15}$$

The other two components give the reaction. Along $\boldsymbol{e}_r$, $-mR\dot\varphi^2 = N_r - mg\sin\varphi + m\Omega^2 x\cos\varphi$, and along $\boldsymbol{k}$, $0 = N_z + m\dot\Omega x - 2m\Omega R\dot\varphi\sin\varphi$. Solving for N_r and N_z,

$$N_r = -mR\dot\varphi^2 + mg\sin\varphi - mA^2\omega^2 R\sin^2(\omega t)(1+\cos\varphi)\cos\varphi, \tag{16}$$

$$N_z = mA\omega R\big[2\dot\varphi\sin(\omega t)\sin\varphi - \omega\cos(\omega t)(1+\cos\varphi)\big]. \tag{17}$$

By the equation obtained, Equation (15), it can be deduced that:

- If the plane is at rest ($A = 0$), the bead is a pendulum, $\ddot\varphi = -(g/R)\cos\varphi$ (cosine and not sine because φ is measured from the horizontal). The stable equilibrium is at the bottom, $\varphi = -\pi/2$, and the unstable one at the top, $\varphi = \pi/2$.
- Only the centrifugal force changes the motion along the wire. The Euler and Coriolis forces are normal to the plane, so they only load the wire through N_z. For the oscillating loop studied here, the centrifugal term is at most $(A\omega)^2/(g/R) \approx 1\%$ of gravity, so we expect the bead to move almost like a pendulum on a fixed loop.

2.3 Mechanical energy

The kinetic energy has to use the velocity seen from the ground, Equation (9):

$$E_k = \tfrac{1}{2}m(R^2\dot\varphi^2 + \Omega^2 x^2), \qquad E_p = mgR\sin\varphi, \qquad E = E_k + E_p. \tag{18}$$

The weight is already included in E, so E can only change because of the work of the reaction. Using Equations (9) and (11) and $\boldsymbol{e}_r \cdot \boldsymbol{e}_\varphi = 0$,

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \boldsymbol{N}\cdot\boldsymbol{v} = (N_r\,\boldsymbol{e}_r + N_z\,\boldsymbol{k})\cdot(R\dot\varphi\,\boldsymbol{e}_\varphi - \Omega x\,\boldsymbol{k}) = -\Omega\,x\,N_z. \tag{19}$$

On a fixed wire the reaction is always perpendicular to the velocity and does no work. Here the wire moves with the plane and pushes the bead along $\boldsymbol{k}$, so it does work whenever $\Omega \neq 0$. With this, E is only conserved for $A = 0$.

2.4 Fixed loop: when does the bead go over the top?

For $A = 0$ the energy is conserved Equation (15) by $\dot\varphi$ we get $\dot\varphi\ddot\varphi + (g/R)\cos\varphi\,\dot\varphi = 0$. By the chain rule, $\frac{\mathrm{d}}{\mathrm{d}t}(\frac{1}{2}\dot\varphi^2) = \dot\varphi\ddot\varphi$ and $\frac{\mathrm{d}}{\mathrm{d}t}\sin\varphi = \cos\varphi\,\dot\varphi$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big[\tfrac{1}{2}\dot\varphi^2 + \frac{g}{R}\sin\varphi\Big] = 0 \quad\Rightarrow\quad \tfrac{1}{2}\dot\varphi^2 + \frac{g}{R}\sin\varphi = \tfrac{1}{2}\dot\varphi_0^2 + \frac{g}{R}\sin\varphi_0, \tag{20}$$

which is $E/(mR^2) =$ constant. The highest potential energy is at the top, $\varphi = \pi/2$, where $E_p = mgR$. If the bead arrives there with some speed left, it goes over and φ keeps growing. If not, it stops before the top and comes back. Imposing $\dot\varphi^2 > 0$ at $\varphi = \pi/2$ in Equation (20), we get the criterion

$$|\dot\varphi_0| > \dot\varphi_c = \sqrt{\frac{2g}{R}(1-\sin\varphi_0)}. \tag{21}$$

If both sides are equal, the bead is on the point $\dot\varphi = \pm\sqrt{(2g/R)(1-\sin\varphi)}$, so it moves up to the top but takes an infinite time to reach it. Below the point φ oscillates between the two angles where $\dot\varphi = 0$, which are given by $\sin\varphi = \sin\varphi_0 + R\dot\varphi_0^2/(2g)$. Table 1 applies the criterion to the nine initial conditions studied.

Table 1: Critical rate of Equation (21) ($g/R = 9.81\,\mathrm{s}^{-2}$) and predicted motion for the nine initial conditions.

φ_0 (rad)	$\dot\varphi_c$ (rad/s)	$\dot\varphi_0 = 0$	$\dot\varphi_0 = 1.5$ rad/s	$\dot\varphi_0 = 3$ rad/s
0	4.429	oscillates	oscillates	oscillates
$\pi/4$	2.397	oscillates	oscillates	goes over the top
$\pi/2$	0	rest at the top (unstable)	goes over the top	goes over the top

2.5 Linearized equation

Now we assume that the bead stays close to the bottom, $\varphi = \delta - \pi/2$ with $|\delta| \ll 1$, and that the rotation is weak, $A^2 \ll g/(R\omega^2)$. Then $\cos\varphi = \cos(\delta - \pi/2) = \sin\delta \approx \delta$, $\sin\varphi = -\cos\delta \approx -1$, $1+\cos\varphi \approx 1+\delta$ and $\ddot\varphi = \ddot\delta$. Substituting in Equation (15),

$$\ddot\delta = -\frac{g}{R}\,\delta + \Omega^2(1+\delta) = -\frac{g}{R}\,\delta + \Omega^2 + \Omega^2\delta. \tag{22}$$

As $\Omega^2 \le A^2\omega^2 \ll g/R$, the term $\Omega^2\delta$ is small compared with $(g/R)\delta$ and we drop it (its effect builds up as a phase lag, see Section 3.5). Using $\sin^2(\omega t) = \frac{1}{2}[1-\cos(2\omega t)]$, we get:

$$\ddot\delta + \omega_n^2\,\delta = \Omega^2 = F_0\big[1-\cos(2\omega t)\big], \qquad \omega_n = \sqrt{g/R}, \qquad F_0 = \tfrac{1}{2}A^2\omega^2. \tag{23}$$

This is an undamped oscillator of natural frequency ω_n, forced at the frequency 2ω [2]. The forcing is the centrifugal force. It is zero when the plane stops and maximum when it turns fastest, which happens twice in each period of the plane. Neither the statement nor the problem sheet [3] gives initial conditions for the linearized study, so we took the bead at rest at the bottom, $\delta(0) = 0$ and $\dot\delta(0) = 0$, so that all its motion comes from the loop. The particular solution is a constant F_0/ω_n^2 plus

$C\cos(2\omega t)$, and substituting the second one, $(\omega_n^2 - 4\omega^2)C = -F_0$. Adding the homogeneous solution $a\cos\omega_n t + b\sin\omega_n t$ and imposing the initial conditions ($\dot\delta(0) = b\,\omega_n = 0$),

$$\delta(t) = \frac{F_0}{\omega_n^2} - \frac{F_0\cos(2\omega t)}{\omega_n^2 - 4\omega^2} + \Big(\frac{F_0}{\omega_n^2 - 4\omega^2} - \frac{F_0}{\omega_n^2}\Big)\cos(\omega_n t), \qquad 2\omega \neq \omega_n. \tag{24}$$

So the bead oscillates about $\delta_s = F_0/\omega_n^2 = A^2 R\omega^2/(2g)$, a bit outwards from the bottom, because the centrifugal force pushes it away from the axis. If $2\omega = \omega_n$, the denominator in Equation (24) is zero. This is resonance, since each push arrives in phase with the natural swing. In that case the particular solution is $-\frac{F_0}{2\omega_n} t\sin(\omega_n t)$, which satisfies $\ddot\delta + \omega_n^2\delta = -F_0\cos(\omega_n t)$, and with the same initial conditions

$$\delta(t) = \frac{F_0}{\omega_n^2}(1 - \cos\omega_n t) - \frac{F_0}{2\omega_n}\, t\sin(\omega_n t), \qquad 2\omega = \omega_n. \tag{25}$$

Now the amplitude grows linearly with time, so at some point $\delta \ll 1$ is no longer true and the linearization stops being valid.

2.6 Numerical method

Since `ode45` only integrates first order systems [4], Equation (15) is written for the state vector $\boldsymbol{X} = [\varphi, \dot\varphi]^T$:

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \begin{pmatrix} X_2 \\ -\dfrac{g}{R}\cos X_1 - \Omega(t)^2\,(1 + \cos X_1)\sin X_1 \end{pmatrix}, \qquad \boldsymbol{X}(0) = \begin{pmatrix}\varphi_0 \\ \dot\varphi_0\end{pmatrix}, \qquad \Omega(t) = A\omega\sin(\omega t). \tag{26}$$

This system is the MATLAB function `diffeq(t,X,g,w,R,A)`, and Equation (23) is `diffeqlin`, with the state $[\delta, \dot\delta]^T$. The tolerances are 10^{-8} (relative) and 10^{-10} (absolute). For the oscillating loop, the solution is stored at 10 000 instants of $[0, 3T]$, which is about 125 points per swing [1]. Forces and energies are computed afterwards from the stored φ and $\dot\varphi$ with exactly Equations (12) to (14) and (16) to (18). The position in the fixed frame comes from Equations (1) and (2), $x_1 = x\cos\theta$, $y_1 = R\sin\varphi$, $z_1 = -x\sin\theta$. In all the cases $g = 9.81\,\mathrm{m/s^2}$ and $m = 1\,\mathrm{kg}$. The oscillating loop uses $A = \pi$, $R = 1\,\mathrm{m}$, $\omega = 0.1\,\mathrm{rad/s}$ and $\varphi_0 = \dot\varphi_0 = 0$ over $[0, 3T]$, with $T = 62.83\,\mathrm{s}$. The fixed loop uses $A = 0$, $R = 1\,\mathrm{m}$ and the nine conditions of Table 1, over 4 s, which is longer than the longest period of the closed curves (3.36 s). The linearized study uses $A = 0.1$, $R = 0.01\,\mathrm{m}$, $\omega = \omega_n,\ \omega_n/2,\ \omega_n/4$ with $\omega_n = 31.32\,\mathrm{rad/s}$, and $\delta_0 = \dot\delta_0 = 0$, over 10 s.

3 Results

3.1 Angle, reaction and inertia forces

At $t = 0$ the plane is at rest, so Equation (15) gives $\ddot\varphi(0) = -g/R = -9.81\,\mathrm{rad/s^2}$, as the code prints. The bead falls from the end of the horizontal diameter and swings between $\varphi = 0$ and $\varphi = -\pi$ (Figure 2a). Over $[0, 3T]$ the 79 swings merge into a solid band and the single swings cannot be seen, so Figure 2b and d repeat φ and N_r for the first 10 s. At $\varphi = -\pi$ it passes through O, on the axis. Each swing takes 2.37 s, so there are 79 swings in $[0, 3T]$. This is longer than the small-oscillation period $2\pi\sqrt{R/g} = 2.006\,\mathrm{s}$, because an amplitude of 90° around the bottom is not small at all. As observed in Figure 2a, the amplitude stays the same over the three periods of the plane, which makes sense since the centrifugal term is only 1% of gravity. The highest angle is $\varphi = 0.0118\,\mathrm{rad}$, so the energy given by the wire lifts the bead about 1 cm above where it started.

The radial reaction (Figure 2c and d) is zero at the turning points and reaches $-29.43\,\mathrm{N}$, which is $-3mg$, at the bottom. There Equation (20) gives $\dot\varphi^2 = 2g/R$, and plugging it into Equation (16), $N_r = -2mg - mg$. The wire holds the weight and also pulls the bead towards C to make it turn. The normal reaction (Figure 2e) reaches $\pm 2.78\,\mathrm{N}$ and has almost the same shape as the Coriolis force (Figure 3c) with the opposite sign, because the wire has to balance it. The maximum of the Coriolis force is $2mA\omega R\dot\varphi_{\max} = 2\pi(0.1)(4.43) = 2.78\,\mathrm{N}$. It is large because the bead moves fast along the wire ($\dot\varphi$ up to 4.4 rad/s) while the plane turns, and it almost vanishes at $t = 0,\ T/2,\ T, \ldots$ ($t = 0,\ 31,\ 63, \ldots$ s), when the plane stops to turn back. In Figure 3 each inertia force has its own axis, since their sizes differ by more than one order of magnitude [1]. The centrifugal force (Figure 3a) always points away

from the axis, and its maximum is $m(A\omega)^2(2R) = 0.197\,\mathrm{N}$, reached when the plane turns fastest, at $t = T/4,\ 3T/4, \ldots$ The Euler force (Figure 3b) follows $\cos\omega t$. It stays below 0.063 N and is largest when the plane stops, because the rate of the plane changes slowly ($|\dot{\Omega}| \leq A\omega^2 = 0.031\,\mathrm{rad/s^2}$).

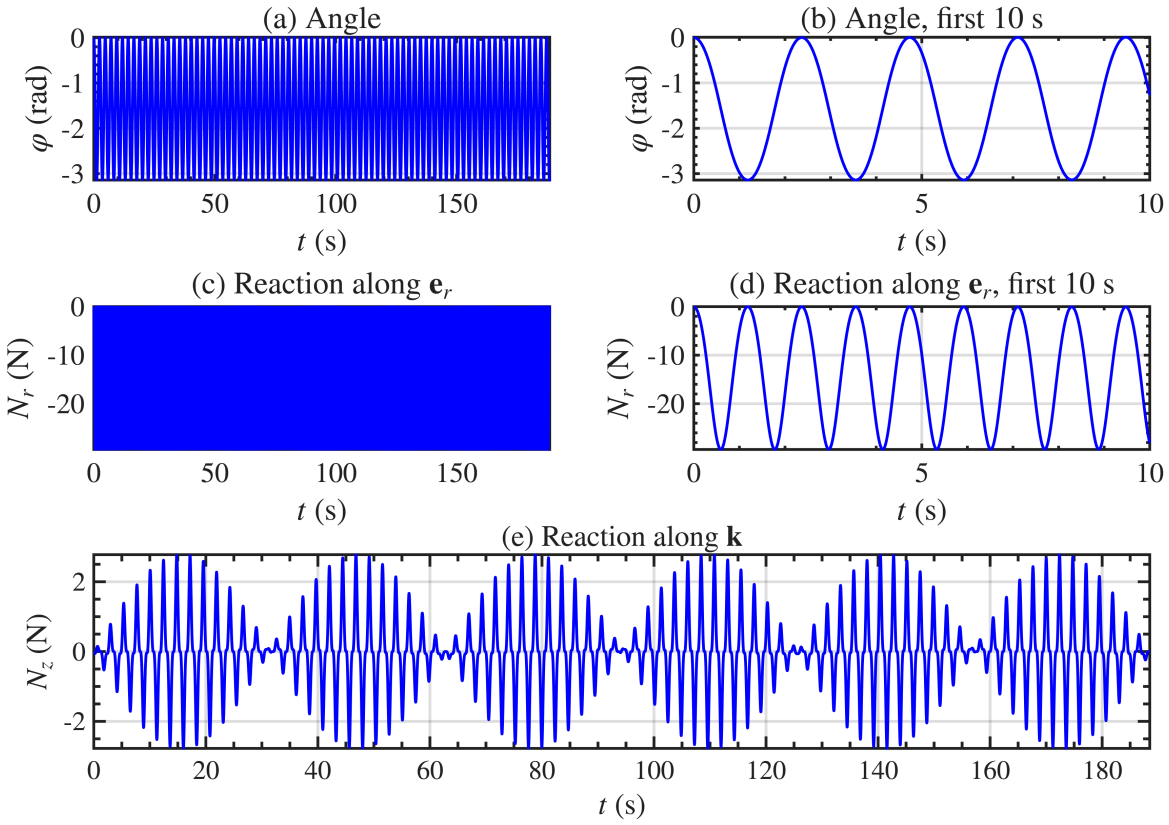


Figure 2: Oscillating loop, $t \in [0, 3T]$: (a) angle, (c) reaction of the wire along $\boldsymbol{e}_r$ and (e) along $\boldsymbol{k}$; (b) and (d) zoom into the first 10 s, where the single swings can be seen.

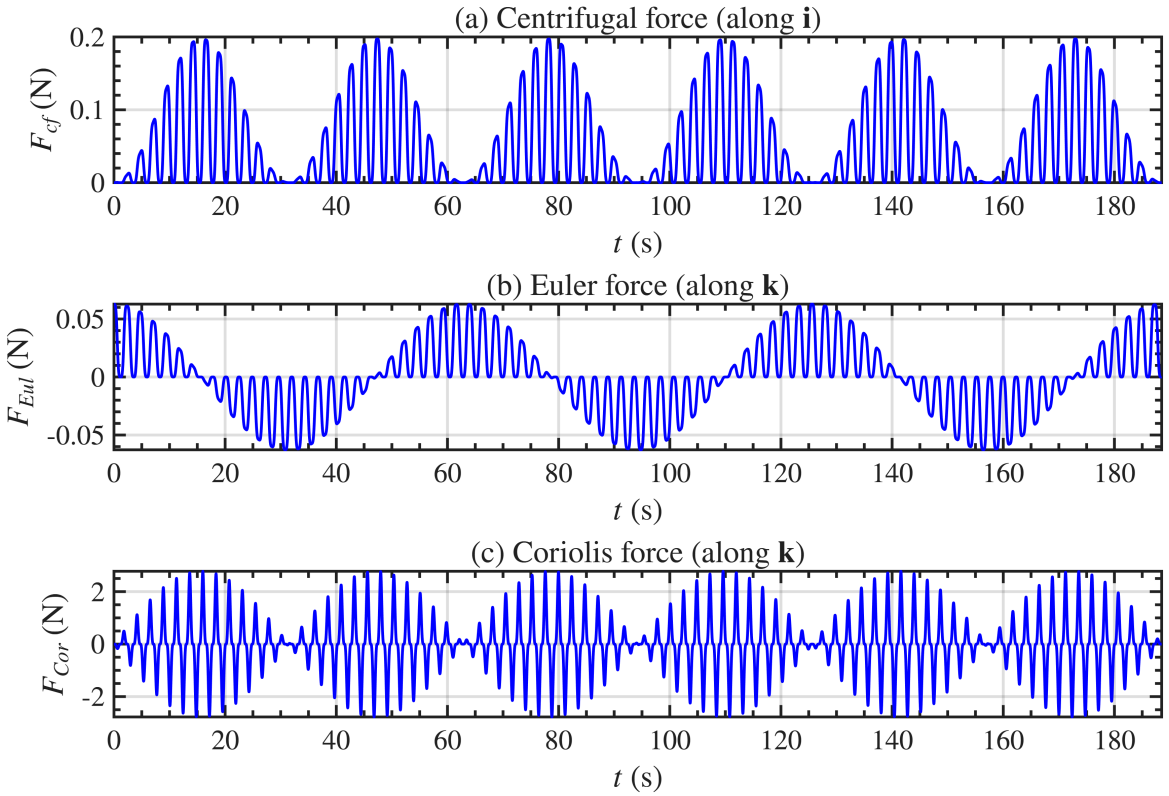


Figure 3: Inertia forces in the turning frame, $t \in [0, 3T]$: (a) centrifugal (along $\boldsymbol{i}$), (b) Euler and (c) Coriolis (along $\boldsymbol{k}$).

3.2 Mechanical energy

In every swing E_k and E_p exchange about 9.8 J, as in any pendulum. Over $[0, 3T]$ (Figure 4a) the two curves also look like solid bands, so Figure 4b shows the first 10 s, where this exchange can be followed swing by swing. Their sum is not constant, but its changes are too small to see at the scale of the figure; the code gives $E - E_0$ between -0.082 J and 0.313 J. While the plane speeds up, the wire pushes the bead with it and gives it energy, and while the plane slows down the wire takes that energy back. The plane changes its speed slowly compared with one swing ($T = 62.8$ s against 2.37 s), so the energy goes back to E_0 every time the plane stops, and after $3T$ the net gain is only 0.0006 J. This explains why the amplitude in Figure 2a does not drift.

With tighter tolerances (10^{-11}, 10^{-12}), φ changes by at most 1.4×10^{-4} rad.

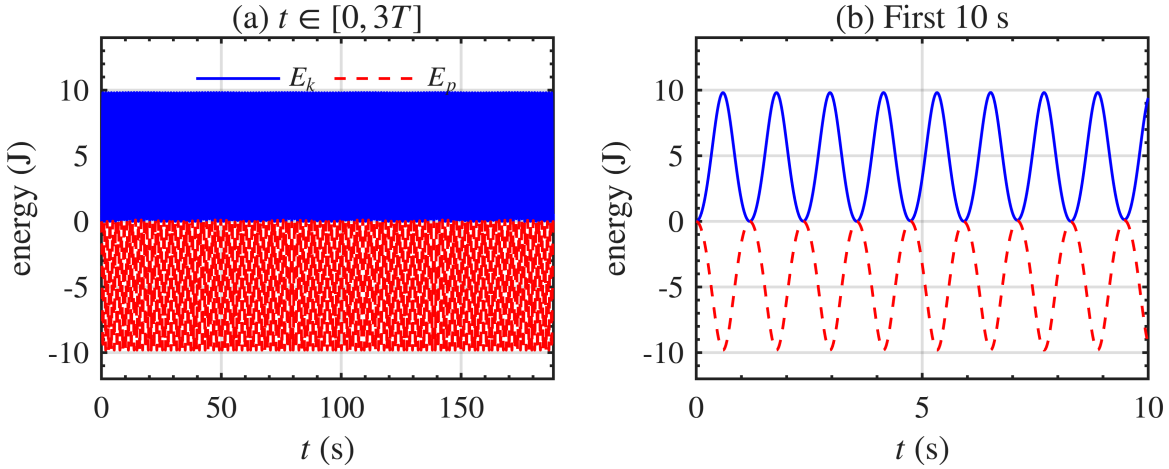


Figure 4: Kinetic and gravitational potential energy of the oscillating loop: (a) $t \in [0, 3T]$ and (b) zoom into the first 10 s.

3.3 Trajectory in the fixed frame

Seen from the ground (Figure 5), the bead always stays on the surface swept by the wire when it turns about Oy. This surface is a torus without a hole, because the circle touches the axis at O. Each swing draws one petal: the bead goes out to $x = 2R$ when $\varphi = 0$ and back to the axis when $\varphi = -\pi$. The petals are spread around the axis because the plane turns up to $\theta = 2\pi$ at $t = T/2$ and goes back to 0 at $t = T$ (Equation (1)). The three periods draw almost the same petals, close to each other but not on top of each other, since 2.37 s does not divide T exactly. As a check, $\sqrt{x_1^2 + z_1^2}$ equals x within 2×10^{-16} m, since a rotation about the axis cannot change the distance to it. The script also saves an animation of the first period, `trajectory_P.avi`, as suggested in the guidelines [1].

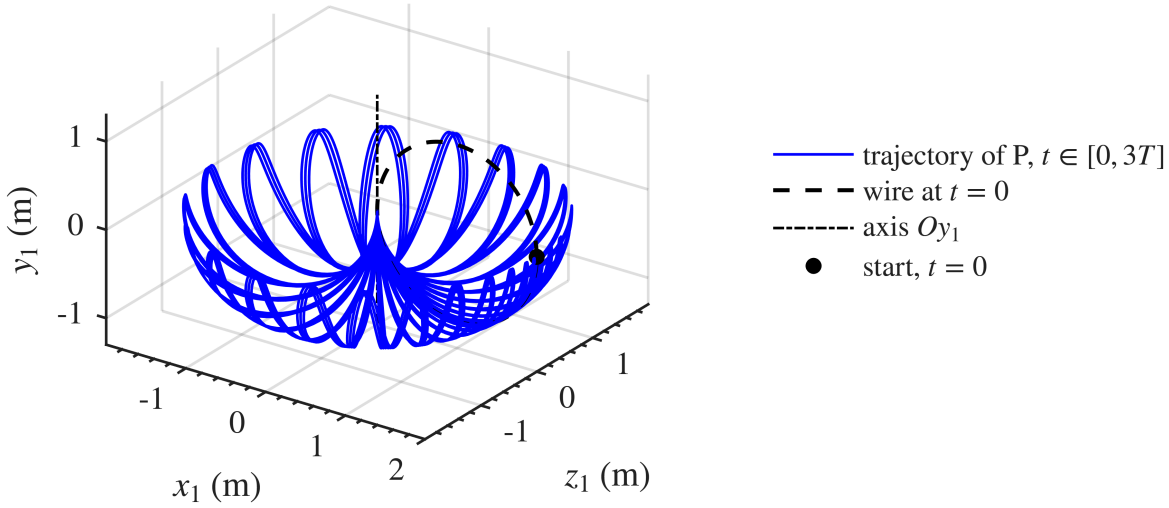


Figure 5: Trajectory of P in the fixed frame $Ox_1y_1z_1$, $t \in [0, 3T]$, with the wire at $t = 0$ and the axis of rotation for reference.

3.4 Phase portrait of the fixed loop

Figure 6 shows the nine cases of Table 1 together with the separatrix of Equation (21). The colour gives φ_0, the line style $\dot{\varphi}_0$, and a dot marks where each curve starts. The five cases $(0, 0)$, $(0, 1.5)$, $(0, 3)$, $(\pi/4, 0)$ and $(\pi/4, 1.5)$ are closed curves inside the separatrix, around the stable equilibrium $(-\pi/2, 0)$, so the bead oscillates. The period goes from 2.37 s to 3.36 s as the curve gets closer to the separatrix, because the bead slows down near the top. The cases $(\pi/4, 3)$, $(\pi/2, 1.5)$ and $(\pi/2, 3)$ start above the separatrix and give open curves: the bead keeps going round, slowest at the top and fastest at the bottom, and the figure is cut at $\varphi = 5\pi/2$, since the curves repeat every 2π. The case $(\pi/2, 0)$ starts on the unstable equilibrium and stays there as a single point, since numerically $\cos(\pi/2) \approx 6 \times 10^{-17}$ is far too small to move it in 4 s. As a conclusion, all nine cases behave as Table 1 predicts, and in all of them Equation (20) holds to better than $2 \times 10^{-7}\, mgR$.

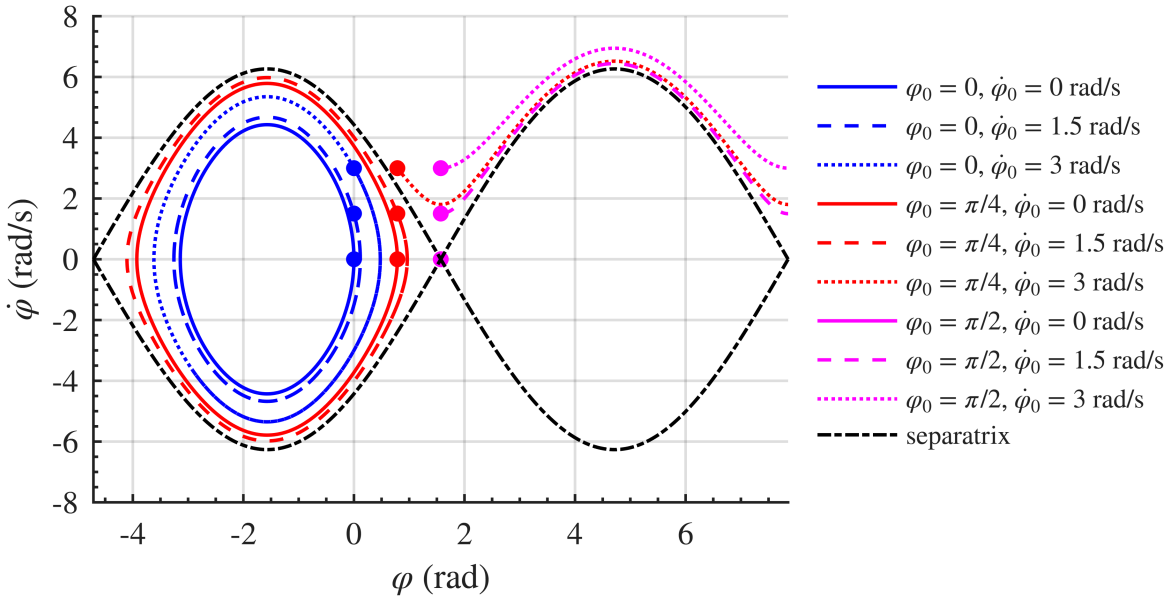


Figure 6: Phase portrait of the fixed loop ($A = 0$), for the nine initial conditions, with the separatrix of Equation (21). The open curves repeat every 2π.

3.5 Linearized against full equation

Figure 7 shows the percentage error $100\,|\delta_{\text{lin}} - \delta_{\text{full}}| / \max|\delta_{\text{full}}|$. We divide by the amplitude of δ and not by δ itself, because δ crosses zero. The error goes from 0.01% to almost 30%, so the vertical axis is logarithmic, as the guidelines suggest [1]. The downward spikes are the instants where both solutions cross. The `ode45` solution of Equation (23) matches Equations (24) and (25) within 2×10^{-8} rad, so the differences come from the model, not from the integration (numbers in Table 2).

Table 2: Linearized against full equation, $t \in [0, 10]$ s, with $\omega_n = \sqrt{g/R}$.

ω	$\max\lvert\delta_{\text{full}}\rvert$ (rad)	error at $t = 5$ s	largest error
ω_n	0.0134	6.68%	21.4%
$\omega_n/2$	0.196	3.30%	27.8%
$\omega_n/4$	0.00083	0.32%	1.86%

For $\omega = \omega_n$ (solid line), δ stays small but the error grows steadily from the start, up to 21%. The cause is the term $\Omega^2\delta$ dropped in Section 2.5, which lowers the stiffness of the full equation from $g/R = 981\,\text{s}^{-2}$ by about $A^2\omega^2/2 = 4.9\,\text{s}^{-2}$ on average. So the full solution oscillates 0.25% slower and falls behind, by about 0.8 rad of phase after 10 s.

For $\omega = \omega_n/2$ (dashed line), $2\omega = \omega_n$, which is the resonance of Equation (25) [2]. Both solutions grow like $F_0 t/(2\omega_n)$, up to 0.196 rad (11°) at 10 s. At first they agree, so the error starts as the smallest of the three, but as δ grows $\sin\delta \approx \delta$ gets worse, the full pendulum slows down and gets out of phase with the forcing, and the error ends as the largest, 27.8%.

For $\omega = \omega_n/4$ (dotted line), the forcing is far from resonance and 16 times weaker than in the first case: δ stays below 9×10^{-4} rad and the error below 2%, so here the linearization works.

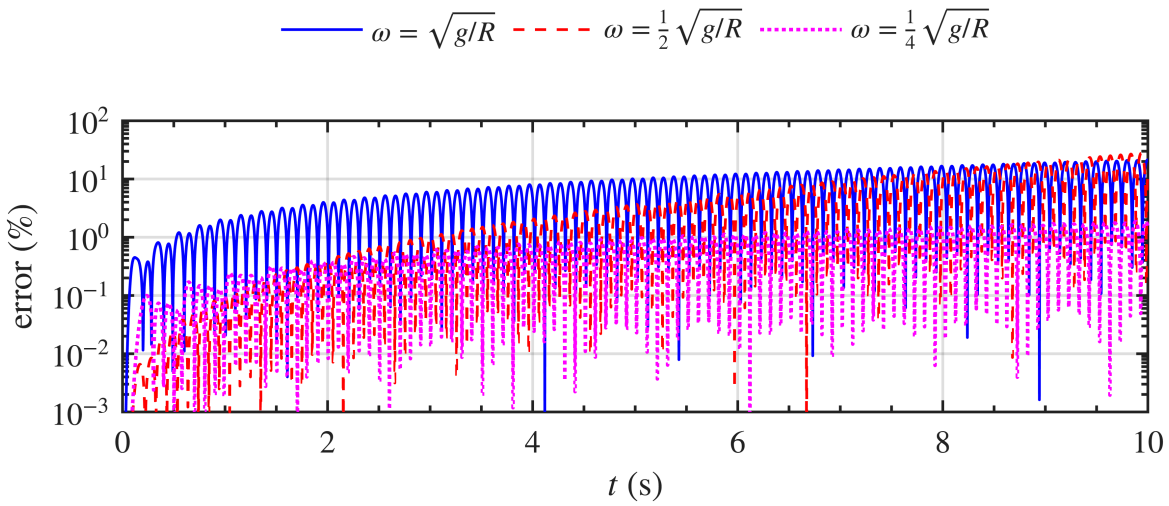


Figure 7: Percentage error between the linearized and the full solutions ($R = 0.01\,\text{m}$, $A = 0.1$), in logarithmic scale.

4 Discussion

The energy results can be explained from two points of view. Seen from the ground, E is not conserved because the wire moves and its reaction does work on the bead (Equation (19)). In the turning frame, the reaction and the Euler and Coriolis forces are perpendicular to $\boldsymbol{v}_{\text{rel}}$, so they do no work. However, the centrifugal potential $-\frac{1}{2}m\Omega^2x^2$ depends on time, and for this reason $\frac{1}{2}mR^2\dot{\varphi}^2 + E_p - \frac{1}{2}m\Omega^2x^2$ is not conserved either. If Ω were constant and different from zero, this energy of the turning frame would be conserved, but E of Equation (18) would still change, since the Coriolis force makes $N_z \neq 0$. This means that E is only conserved for $A = 0$, which is the case of the fixed loop.

Some choices of the model also affect the results. In the linearized study the bead starts at rest at the bottom; with $\delta(0) \neq 0$ a free oscillation would be added, which would change the size of the errors but not the phase lag or the resonance. The model also assumes a smooth wire and no damping, so no energy is taken out of the bead. This is why, at resonance, the linearized solution grows without bound, while a real bead would probably reach a finite amplitude because of friction.

5 Conclusions

The bead has a single degree of freedom, φ, and in the turning frame its equation of motion only contains gravity and the centrifugal force. On the oscillating loop the bead swings like a pendulum, with an amplitude of 90° and a period of 2.37 s. The wire has to provide up to $3mg$ in its plane and up to 2.78 N normal to it, and this normal component mostly balances the Coriolis force. Since this push does work, the energy is not conserved, although the work cancels out every time the plane stops.

When the loop is fixed, energy conservation gives the critical rate $\dot{\varphi}_c = \sqrt{(2g/R)(1 - \sin\varphi_0)}$, and the nine simulated cases agree with it. The linearized equation behaves as a forced oscillator. It works well for weak forcing far from resonance, it drifts in phase because of the dropped term $\Omega^2\delta$, and it fails at resonance ($\omega = \omega_n/2$), where δ keeps growing.

As a conclusion, `ode45` was enough to study a particle on a moving constraint, and the numerical results agree with the analytical ones: the initial acceleration $-g/R$, the reaction $-3mg$ at the bottom, the critical rate of Equation (21) and the closed forms of the linearized equation. As a future extension, the full nonlinear equation could be integrated for longer than 10 s near resonance to see how the nonlinear terms limit the growth of δ.

References

[1] UC3M Aerospace Engineering, *Mechanics Applied to Aerospace Engineering: Laboratory Guidelines*, 2026.

[2] UC3M Aerospace Engineering, *Mechanics Applied to Aerospace Engineering: course notes*, Sec. 5.4.

[3] UC3M Aerospace Engineering, *Particle exercises and problems*, Problem 34.

[4] The MathWorks, Inc., *MATLAB documentation: ode45, odeset, VideoWriter, print.*